

Diskmath Week 3

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Hello

I am Emil.

I have a **website**:

emils.site (yes thats the URL)

repo: <https://github.com/emil3tr/emil3tr.github.io>

All materials will be uploaded there.

Any Feedback / Questions /
Wishes / ... ?

→ estraschil@student.ethz.ch

→ “Emil” (floxi4) on dinfk-discord

→ Diskmath-questions: ask here
so others can benefit (:

Where are we right now?

Basics

Abstraction
Formulas
Statements
Prop. Logic
Pred. Logic
→ Proof Patterns

Sets and
Relations

Sets
Set Operations
Relations
Equivalence
Partial Order
Functions
Countability

Number Theory

Division
Primes
Modular Arith.
Diffie-Hellman

Algebra

Monoids
Groups
Euler Totient
RSA
Rings
Polynomials
Finite Fields
Err. Corr. Codes

Logic

Proof Systems
Logic
Calculi
Res. Calculus
Prop. Logic
Pred. Logic

Today

We have a lot to do...

1. Questions?
2. Last Serie
3. Recap Logic
4. Kahoot
5. PROOF PATTERNS DEEP DIVE
6. Exercise
7. (Sets?)
8. Outlook

I have some slides for you (:

Questions?

Logic Recap

What is important?

- Predicate Logic
- Lemma 2.1
- Quantifiers
- Quantifier Rules

You will see all of this again in chapter 6.

Predicates

Functions that either evaluate to true or false.

Allow us to “embed” non-logic into logic.

Example: $\text{prime}(x)$ is true only when x is a prime.

The Universe

Set of all elements that the formula
“works on”.

Quantifiers

$\forall$ means “for all”

$\exists$ means “exists”

Interpretations

An interpretation of a formula fixes:

1. The universe U
2. The meaning of the predicates
3. Other “free” symbols (more in chapter 6)

Example

$\forall x \exists y P(x,y)$ is a **formula**.

→ Interpretation:

- Universe are the natural numbers
- $P(x,y)$ is true exactly if y is greater than x

Now it is a statement!

Formulas from Sentences

Use the quantifiers and prop. logic to express the sentence as formula

Tips:

- “if x is A then x must also be B” becomes $\text{isA}(x) \rightarrow \text{isB}(x)$
- “y is the only A” becomes $\forall x (x=y \vee \neg \text{isA}(x))$

Example: “every even number is divisible by 2” becomes:

$\forall x (\text{even}(x) \rightarrow \text{divides}(2,x))$

Formulas from Sentences

Look at exercises from last week
for examples!

<https://discmath.ch/content/ch2/predicate-logic> also has examples
(scroll down to last example)

Kahoot

Proof Patterns

General

- Proof patterns are very important
- You will need them in all further topics
- Proofs may seem arbitrary at first, but there are patterns.
- Read through all solutions, note down good ideas and properties used
- The only way to learn proofs is PRACTICE

Modus Ponens

- We show A
- We show A implies B
- Now we have shown B

- Do this multiple times to get

$A \rightarrow B \rightarrow \dots \rightarrow S$

- Most parts of your proofs will look like this

Proving implications ($A \Rightarrow B$)

Directly

- Assume A
- Show B

Indirectly

- Assume not B
- Show not A

$(A \Rightarrow B)$ same as

$(\text{not } A \text{ or } B)$ same as

$(\text{not not } B \text{ or not } A)$ same as

$(\text{not } B \Rightarrow \text{not } A)$

Case Distinction

Want to show statement with different cases.

Example: Number odd or even

- Prove for each case individually
- Each case works and one needs to hold, so the statement holds

Proof by Contradiction

Want to show statement S.

Explanation weird in script, this is how we do it:

- Start by assuming S is false
- Show T from that assumption
- Show that T is false

Exhibiting Example

Want to show something with 'exists'.

- Simply give an example

Counterexample

Want to show 'not for all'.

- Simply give counterexample

Don't forget to show WHY it is a counterexample. Simply stating it is not enough.

(Except if exercise says otherwise.)

Counterexamples in Exams

Some exercises will be
“prove or disprove”

Counterexamples are the easiest
points in the exam.

- Search for counterexamples first
- Try 0, 1, -1, empty set, edge cases, intuition, ...
- Only start to prove when you understand why it is true

Pigeonhole Principle

If we have n pigeons and less than n boxes, two pigeons have to go in the same box.

Induction

Want to show something for all natural numbers.

(Or for all numbers from x , $x+1$, ...)

- Show it for 0
- Assume it holds for n
- Show it for $n+1$

Domino principle.

All sheep have the same
color

Correctness of Proof Patterns

Use Propositional Logic

For each statement in the pattern
S, T, ... choose a corresponding
proposition symbol A, B, ...

“A is set to true if S is true”

...

Transform assumptions to
formulas and show that the
logically imply the one you want
to prove.

See

[https://emils.site/docs/diskmath
_hs25/sessions/week3/](https://emils.site/docs/diskmath_hs25/sessions/week3/)

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2023 3.2